

Question 1

- a) Let $U = \{x \mid x \text{ is a positive integer and } x \leq 10\}$ be the universal set that contains the following subsets:

$$A = \{x \mid x \text{ is a prime number and } x < 13\},$$

$$B = \{x \mid x \text{ is a multiple of 3}\} \text{ and}$$

$$C = \{x \mid x^2 - 7x + 10 = 0\}.$$

- (i) List all the elements of U , A , B and C . (4 marks)
- (ii) Find $\overline{B} - C$. (2 marks)
- (iii) Find $A \oplus C$. (2 marks)

(i) $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

$$A = \{2, 3, 5, 7\}$$

$$B = \{3, 6, 9\}$$

$$C = \{2, 5\}$$

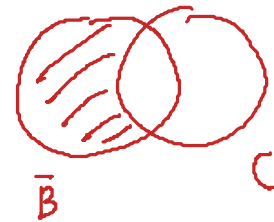
(ii) $\overline{B} = \{1, 2, 4, 5, 7, 8, 10\}$

$$\begin{aligned}\overline{B} - C &= \{1, 2, 4, 5, 7, 8, 10\} - \{2, 5\} \\ &= \{1, 4, 7, 8, 10\}\end{aligned}$$

(iii) $A \oplus C = (A \cup C) - (A \cap C)$

$$\begin{aligned}&= \{2, 3, 5, 7\} - \{2, 5\} \\ &= \{3, 7\}\end{aligned}$$

$$\overline{B} - C$$



$$A \oplus C$$



- b) By using the Euclidean algorithm, find the greatest common divisor (GCD) of 1140 and 672. Hence, determine the least common multiple (LCM) of 1140 and 672. (5 marks)

b) $1140 = 672(1) + 468$

$$672 = 468(1) + 204$$

$$468 = 204(2) + 60$$

$$204 = 60(3) + 24$$

$$60 = 24(2) + 12$$

$$24 = 12(2) + 0$$

$$\gcd(1140, 672) = 12$$

$$\text{lcm}(1140, 672) = \frac{1140 \times 672}{12} = 63840$$

$$\begin{array}{r} 1 \\ 672 \overline{) 1140} \\ \underline{672} \\ 468 \end{array}$$

a) Let R be a relation on the set $A = \{0, 1, 4, 8\}$ defined by xRy if and only if $x + y < 8$.

(i) List the ordered pairs which belong to the relation R . (2 marks)

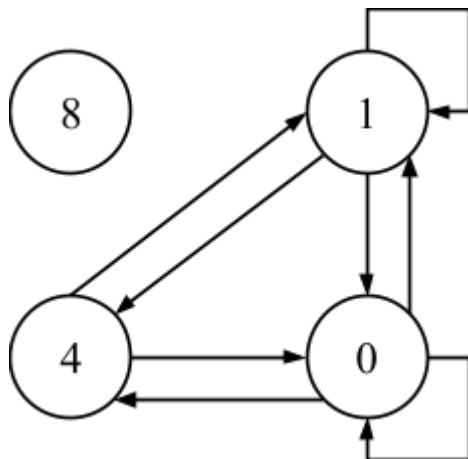
(ii) Draw the digraph of R . (2 marks)

(iii) Represent R in matrix form. (2 marks)

(iv) Determine the in-degree and out-degree of each vertex. (2 marks)

(i) $R = \{(0, 0), (0, 1), (0, 4), (1, 0), (1, 1), (1, 4), (4, 0), (4, 1)\}$

(ii)



(iii)

$$M_R = \begin{pmatrix} 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 4 & 1 & 1 & 0 & 0 \\ 8 & 0 & 0 & 0 & 0 \end{pmatrix}$$

(iv)

Vertex	0	1	4	8
In-degree	3	3	2	0
Out-degree	3	3	2	0

Question 2

a) Let $A = \{a, b, c, d, e\}$ and let $R = \{(a, a), (b, b), (c, c), (d, d), (e, e), (a, b), (b, a), (b, c), (c, d), (d, e), (a, e)\}$ be a relation defined on A .

(iii) Determine whether the relation R is reflexive, irreflexive, symmetric, asymmetric, antisymmetric or transitive. Give a counterexample if the answer is "No". (11 marks)

R is reflexive.

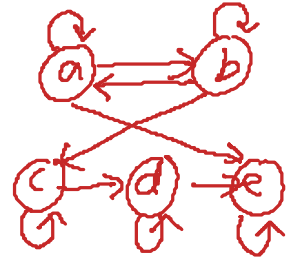
R is not irreflexive since $(a, a) \in R$.

R is not symmetric since $(b, c) \in R$ but $(c, b) \notin R$.

R is not asymmetric since $(a, a) \in R$.

R is not antisymmetric since $(a, b) \in R$ and $(b, a) \in R$.

R is not transitive since $(a, b) \in R$ and $(b, c) \in R$ but $(a, c) \notin R$.



a) Let $A = [p \vee \sim (q \leftrightarrow r)] \wedge (q \rightarrow \sim r)$.

- (i) Construct a truth table for A. Determine whether the expression is a tautology, contradiction or contingency. (6 marks)
- (ii) Find the Principal Disjunctive Normal Form (PDNF) and Principal Conjunctive Normal Form (PCNF) of A and $\sim A$. (4 marks)

p	q	r	$q \leftrightarrow r$	$\sim(q \leftrightarrow r)$	$p \vee \sim(q \leftrightarrow r)$	$\sim r$	$q \rightarrow \sim r$	A
0	0	0	1	0	0	1	1	0
0	0	1	0	1	1	0	1	1
0	1	0	0	1	1	1	1	1
0	1	1	1	0	0	0	0	0
1	0	0	1	0	1	1	1	1
1	0	1	0	1	1	0	1	1
1	1	0	0	1	1	1	1	1
1	1	1	1	0	1	0	0	0

A is a contingency.

(ii)

PDNF of A = $\bar{p}\bar{q}r + \bar{p}q\bar{r} + p\bar{q}\bar{r} + p\bar{q}r + pq\bar{r}$

PDNF of $\sim A$ = $\bar{p}\bar{q}\bar{r} + \bar{p}qr + pqr$

PCNF of A = $(p + q + r)(p + \bar{q} + \bar{r})(\bar{p} + \bar{q} + \bar{r})$

PCNF of $\sim A$ =

$(p + q + \bar{r})(p + \bar{q} + r)(\bar{p} + q + r)(\bar{p} + q + \bar{r})(\bar{p} + \bar{q} + r)$

- b) Write the converse, inverse, contrapositive and negation statements for the following statement:

$\overset{P}{\text{If I exercise regularly,}} \text{ then } \overset{Q}{\text{I stay healthy.}}$

(4 marks)

$$Q \rightarrow P$$

Converse: If I stay healthy, then I exercise regularly.

$$\sim P \rightarrow \sim Q$$

Inverse: If I don't exercise regularly, then I don't stay healthy.

$$\sim Q \rightarrow \sim P$$

Contrapositive: If I don't stay healthy, then I don't exercise regularly.

Negation: I exercise regularly but I don't stay healthy.

$$\sim (P \rightarrow Q) \quad \text{and}$$

$$= \sim (\sim P \vee Q) = P \wedge \sim Q$$